Reza M. Asiyabi

Machine Learning Scientist · Geospatial AI · Earth Observation

Right to Work: **UK Global Talent Visa** (no sponsorship required in the UK)

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SUMMARY

Geospatial AI/ML | Applied AI for Geospatial/Environmental Intelligence | GeoFoundation Models

Machine learning and remote sensing scientist building applied AI systems for large-scale Earth observation data, from model development through to deployment. Currently, a Machine Learning Scientist at Treefera, building geospatial AI for supply chain and environmental intelligence; previously led ML/EO projects at the University of Edinburgh, the UK's National Centre for Earth Observation (NCEO), Stanford University, and ESA-funded initiatives, developing computer vision and deep learning models for multimodal satellite data (SAR, optical, LiDAR) and geospatial time series. Background spans representation learning, interpretable models, and physics-aware AI. Comfortable working across technical and commercial teams to move ambiguous problems into deployed tools, with strong Python and large-scale experimentation skills.

CORE TECHNICAL EXPERTISE AND SKILLS

Remote sensing data	SAR, Multispectral, LiDAR, hyperspectral; sensor fusion and multi-modal geospatial analysis
EO processing tools	Google Earth Engine, GDAL, Geopandas, Rasterio, Xarray, TerraTorch, TorchGeo, SNAP, ENVI, QGIS, ArcGIS
Application domains	Environmental monitoring, land cover mapping, vegetation/crop health monitoring, change detection
Geospatial ML	Geo-Foundation models, domain-aware and physics-informed learning, large-scale geospatial data pipelines
Frameworks	PyTorch, JAX, TensorFlow/Keras, HuggingFace Transformers
Architectures	CNNs, Vision Transformers, LSTMs, Autoencoders, Complex-Valued Networks, Concept Bottleneck Models
Specialisations	Foundation models, self-supervised learning, multimodal fusion, few-shot learning, physics-informed neural networks, neural data compression, explainable AI (XAI)
MLOps	End-to-end training pipelines (HPC & cloud), CI/CD, experiment tracking (TensorBoard, W&B), model deployment, version control (Git), proof-of-concept to production
Python ecosystem	NumPy, SciPy, Pandas, Scikit-Learn, OpenCV, Matplotlib
Large-scale pipelines	Distributed data processing, cloud-based geospatial workflows, high-volume EO dataset management
Other	GCS, AWS, R, MATLAB, SQL, C++ (working knowledge)
Track record	20+ publications, 700+ citations; IEEE TGRS, J-STARS, J-STSP; cross-functional collaboration with academic and industry partners
Languages	English (fluent), Persian/Farsi (native), Turkish/Azari (native)

PROFESSIONAL EXPERIENCE

Machine Learning Scientist – Geospatial AI for Supply Chain Intelligence
Oct 2026 - Present

Treefera · London, UK

AI-powered data fabric fusing spatial and temporal insights to deliver first-mile visibility, sourcing, risk and compliance intelligence for resilient supply chains.

- Developing robust, interpretable machine learning models for complex, high-dimensional spatial and temporal data, supporting market, risk and environmental intelligence.
- Applying deep learning, computer vision and multimodal learning across radar, multispectral and LiDAR data.

Postdoctoral Research Associate (PDRA) – AI for Environmental Monitoring
Aug 2024 - Oct 2026

The University of Edinburgh – National Centre for Earth Observation (NCEO) · Edinburgh, UK

Developed domain-aware AI systems for environmental monitoring using multimodal and noisy Earth Observation data.

- Designed and implemented novel modular architectures (PG-CBM and StruMPL) for vegetation monitoring, integrating domain physics constraints with interpretable concept/multi-task learning to reduce prediction bias by 80% and improved ecological alignment.
- Built multi-stage learning pipeline with data augmentation and regularization strategies, combining multi-modal satellite and field/plot data with physics-based loss functions to enforce structural constraints, improve model robustness against label noise, MNAR bias, domain shift, and spurious correlations in sparse-label regimes.
- Collaborated with domain scientists to design interpretable model outputs aligned with ecological theory.

Postdoctoral Research Assistant – Remote Sensing for Economic Monitoring
Jan 2024 to Jul 2024

Stanford University / Stockholm School of Economics · Remote

Market Activity Index (MAI) project, Center for Food Security and the Environment (FSE).

- Designed novel remote sensing time-series change detection methods to detect and track economic activity of local markets.
- Implemented processing pipeline for satellite time-series imagery on Google Earth Engine and Google Cloud platform.

AI Research Engineer – Neural Data Compression
Jan 2022 to Jan 2023

European Space Agency (ESA) funded project – ARTISTE · Adaptive SAR Signal Compression Through Artificial Intelligence

Led AI development for neural compression of raw SAR data for next-generation ESA satellite missions.

- Architected autoencoder-based compression pipeline using complex-valued neural networks to preserve phase coherency in SAR raw data. Achieved 8:1 compression ratio while maintaining signal fidelity for downstream processing.
- Designed custom loss functions balancing reconstruction accuracy with information-theoretic compression objectives.
- Implemented end-to-end training and evaluation framework and validated against ESA missions' operational SAR processing chains.
- Collaborated with engineering teams at DLR and Airbus to integrate ML compression module into mission simulation environment and assess computational feasibility for onboard processing.

Visiting Researcher – Complex-valued Deep Learning
Oct 2021 to Oct 2022

Zentrum für Sensorsysteme (ZESS), University of Siegen · Siegen, Germany

Research collaboration focused on theoretical and applied development of complex-valued neural networks.

- Designed novel complex-valued convolutional architectures preserving phase information in SAR imagery. Demonstrated ~20% accuracy improvement over real-valued baselines on classification tasks.
- Developed theoretical framework for backpropagation and gradient computation in complex-valued networks, implemented custom PyTorch layers for complex activations and operations.

PhD Research Assistant – Deep Learning for Earth Observation
Dec 2020 to Dec 2023

CEOSpaceTech, University Politehnica Bucharest · Bucharest, Romania

Marie Sklodowska-Curie Early Stage Researcher (EU Horizon 2020 fellowship) conducting fundamental research on robust deep learning architectures for multimodal Earth Observation data.

- Developed end-to-end deep learning framework for complex-valued SAR data processing, combining semantic segmentation, classification, and neural compression in unified architectures.
- Designed coherency-preserving neural networks maintaining physical signal properties through complex-valued operations.
- Created and released S1SLC_CVDL: annotated complex-valued SAR dataset enabling reproducible research in complex-valued networks.

SELECTED SIDE PROJECTS (more details on my GitHub)

Deep Learning for BioMassters	**AGB Assistant LLM**	**Foundation Model Fine-Tuning**
Geospatial Foundation Model for AGB	**EuroSAT Land Cover Classification**	**Geospatial Foundation Model Benchmark**

EDUCATION

PhD of Electronics, Telecommunications and Information Technology
Dec 2020 to Dec 2023

University POLITEHNICA Bucharest (UPB) · Romania

- Early-Stage Marie Sklodowska-Curie Researcher, EU Horizon 2020 MENELAOS-NT ITN
- Dissertation: "Deep Learning for SAR Data in Presence of Adversarial Samples"

Master of Science (MSc) of Remote Sensing Engineering
Sep 2016 to Sep 2018

K.N. Toosi University of Technology · Tehran, Iran

- Thesis: Bag of Visual Words Model Enhancement for PolSAR Images Classification

Bachelor of Science (BSc) of Geodesy and Geomatics
Sep 2012 to Sep 2016

K.N. Toosi University of Technology · Tehran, Iran

FELLOWSHIPS, AWARDS, AND CERTIFICATES

- Fellow of the University of Edinburgh's **Generative AI Laboratory (GAIL)** (2025-2026)
- **Marie Sklodowska-Curie Doctoral Fellowship**, EU Horizon 2020 (2020-2023), MENELAOS-NT project, Grant No. 860370
- **National Academic Excellence Fellowships**, K.N. Toosi University of Technology, Top 1% nationwide university entrance exams (BSc 2012, MSc 2016)
- Certificate of the IEEE GRSS **High-Performance and Disruptive Computing** in Remote Sensing (2023)

SELECTED PUBLICATIONS (600+ citations on Google Scholar)

WORKING PAPERS

- **R. M. Asiyabi,** et al., "StruMPL: Multi-task Dense Regression under Disjoint Partial Supervision and MNAR Labels" (under review at TMLR – available on arXiv).
- **R. M. Asiyabi,** et al., "PG-CBM: Process-Guided Concept Bottleneck Models" (under review at IEEE TPAMI – available on arXiv).

PUBLISHED DATASETS

- **R. M. Asiyabi,** et al., "S1SLC_CVDL: A Complex-Valued Annotated Single Look Complex Sentinel-1 SAR dataset for Complex-Valued Deep Networks", IEEE DataPort, (2023), doi: 10.21227/nm4g-yd98.

JOURNAL AND CONFERENCE PAPERS

- **R. M. Asiyabi,** et al., "Teaching AI to Think Like an Ecologist: Process-Guided Biomass Estimation from Earth Observation Data", EurIPS 2025 – Workshop on AI for Climate and Conservation, (2025).
- **R. M. Asiyabi,** et al., "Complex-Valued Autoencoder-Based Neural Data Compression for SAR Raw Data," IEEE J-STSP, (2025).
- **R. M. Asiyabi,** et al., "Complex-Valued End-to-End Deep Network with Coherency Preservation for Complex-Valued SAR Data Reconstruction and Classification," IEEE TGRS, (2023).

PRESENTATIONS

- AI/ML/EO sessions at EurIPS 2025, ESA Living Planet Symposium 2022 & 2025, ESA Phi-lab 2021, IEEE IGARSS 2021–2024, EUSAR 2022 & 2024, CBMI 2023, and more.